

Joseph Telaak

704-351-7396 | jttelaak04@gmail.com | [linkedin.com/in/jtelaak](https://www.linkedin.com/in/jtelaak)

EDUCATION

University of South Carolina

BSE in Computer Engineering, Leadership Distinction in Research (GPA: 3.61)

Columbia, SC

Aug. 2022 – Dec. 2024

EXPERIENCE

Chief Executive Officer

Dec. 2024 - Present

Parkeze Inc.

Columbia, SC

- Led end-to-end development of ultra-low-power IoT parking sensors, from concept to market launch.
- Architected cloud-native backend infrastructure supporting real-time data streaming to 1000+ users.
- Optimized sensor hardware and firmware achieving 120% power reduction and 300% cost savings.
- Built comprehensive analytics platform using React, GraphQL, and TimescaleDB for data insights.
- Implemented a highly extensible event ingestion and emission pipeline using Kafka to support 3rd-party systems.
- Developed several extensions for Keycloak's open-source IAM to support our unique authorization requirements.
- Created a CI/CD pipeline to perform end-to-end integration testing and blue/green deployment on Kubernetes.
- Ported the LoRaWAN gateway stack to Zephyr RTOS with custom high-availability features for our gateways.
- Involved in all aspects of the business, including fundraising, sales, marketing, and customer support.

Research Assistant

Feb. 2023 - Dec. 2024

USC Systems Research on X Lab

Columbia, SC

- Co-authored research on novel mmWave radar algorithms for 3D vehicle and pedestrian detection.
- Secured research funding through competitive grant proposals.
- Developed contactless vital sign monitoring system.
- Designed data collection infrastructure supporting multiple research projects.

Project Lead

Jan. 2022 - Feb. 2023

SCGSSM Autonomous Golf Cart

Hartsville, SC

- Transformed a standard golf cart into an autonomous vehicle platform for research.
- Engineered a complete ADAS control module for the golf cart's steering, throttle, and braking systems.
- Designed custom Nvidia Jetson carrier board with multi-channel analog-to-digital video capture capabilities.
- Implemented computer vision pipeline for obstacle detection, traffic sign recognition, and lane tracking.

LEADERSHIP

FIRST Robotics

Columbia, SC

FIRST Technical Advisor, Judge, Robot Inspector

Jan. 2022 - May 2025

SELECTED PROJECTS

PiSwitch: Created a 7-port router/network switch with custom controller, web UI, and out-of-band management.

LLM Voice Assistant: Designed a far-field mic array with XMOS DSP, Zephyr on i.MXRT, ESP32 Wi-Fi NIC, OTA for all 3 major chips, and real-time audio streaming to a LLM pipeline.

Agentic Audio Streamer: Developed a multi-room audio streaming system using an SigmaDSP, DACs, and Raspberry Pi CM5. Integrated Spotify Connect and enabled external control via a custom API for LLMs.

RISCv CPU: Implemented a parallelized RISCv CPU with GPIO on an FPGA.

Pick-n-Place Machine: Designed and built a desktop PCB assembler running Marlin and OpenPNP.

SKILLS

Software: C/C++, Python, Java, MATLAB, Rust, LUA

Tooling: Altium, KiCad, Quartus, STM32 Cube, MXUExpresso, mmWave Studio

Embedded: Zephyr, Yocto, Buildroot

Memberships: IEEE Eta Kappa Nu (HKN), IEEE MTTs, IEEE, ACM, AIAA